

UNIT 10

BIOCHEMISTRY



What is biochemistry?

Definition: Biochemistry is the study of biomolecules or the chemical processes that happens inside living organisms.

CARBOHYDRATES:

DEFINITION: Carbohydrates are a class of essential biomolecules found in living organisms, and they serve as one of the main sources of energy for the body.

ELEMENTS: Chemically, carbohydrates are composed of carbon, hydrogen, and oxygen atoms, with a general formula of $(CH_2O)_n$, where "n" represents the number of repeating units.

Structural analysis

Polyhydroxy aldehydes: They contain aldehyde group ($-CHO$) with alcohol ($-OH$).

Polyhydroxy ketones: They contain ketonic group ($-CO-$) with alcohol ($-OH$).

Classification:

Carbohydrates can be classified into three main groups based on the number of sugar units they contain:

1. MONOSACCHARIDES

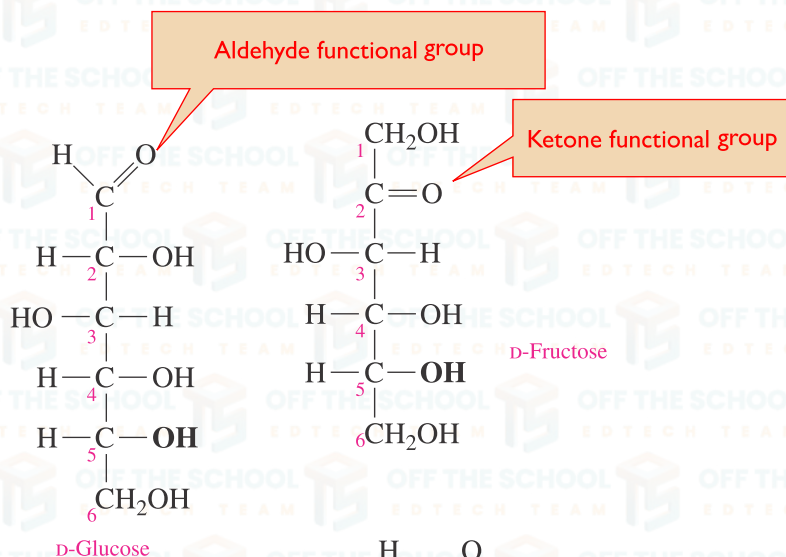
Monosaccharides, also known as simple sugars, are a type of carbohydrate that cannot be further broken down through hydrolysis.

They consist of 3 to 10 carbon atoms and can be classified into different groups such as trioses, tetroses, pentoses, and hexoses, depending on the number of carbon atoms they contain.

Monosaccharide + water $\longrightarrow$ **No reaction**



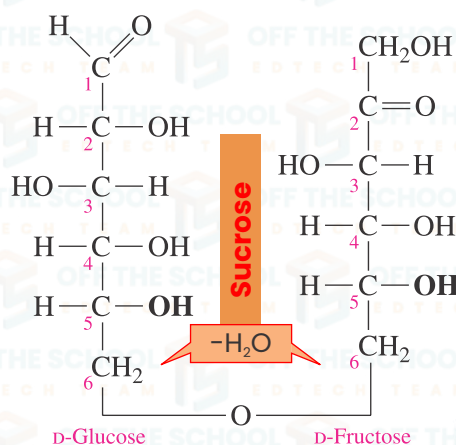
The sweet taste of honey is due to the monosaccharides D-glucose and D-fructose.



2. OLIGOSACCHARIDES

Oligosaccharides are a type of carbohydrate composed of a small number of monosaccharide units (typically 3 to 10) linked together through glycosidic bonds.

Oligosaccharides + water $\xrightarrow{\text{Hydrolysis}}$ **2-10 monosaccharides**



3. POLYSACCHARIDES: Polysaccharides are complex carbohydrates composed of multiple monosaccharide units linked together in long chains. They serve as storage forms of energy in plants and animals and often have structural roles. Common examples of polysaccharides include starch (energy storage in plants), glycogen (energy storage in animals), and cellulose (structural component in plant cell walls).



Importance/Uses of Carbohydrates:

1. Carbohydrates serve as a primary source of energy for the body.
2. Carbohydrates contribute to the structure of cells and tissues, such as cellulose in plant cell walls.
3. Carbohydrates are stored in the form of glycogen in the liver and muscles for later use.
4. Glucose, a carbohydrate, is the preferred energy source for the brain and central nervous system.
5. Carbohydrates play a role in regulating metabolism and maintaining blood glucose levels.

Impact of Carbohydrates on Health and Disease Management:

Glucose:

- A monosaccharide, fundamental to energy production.
- Present in bloodstream; regulated by insulin.
- Primary energy source for cells.
- Overconsumption linked to diabetes, weight gain.
- Maintained in blood at:
 - ✓ Fasting levels: 70–99 mg/dL (normal).
 - ✓ Post-meal (2 hrs): Less than 140 mg/dL (normal).
- Levels above these may indicate hyperglycemia, a symptom of diabetes.

Fructose:

- Simple sugar in fruits, honey.
- Sweetest carbohydrate.
- Provides energy.
- Excess can lead to obesity.

Lactose:

- Disaccharide, "milk sugar".
- Composed of glucose and galactose.
- Difficult for some adults to digest due to enzyme deficiency.
- Essential for infant growth.
- Lactose content: Human milk (5–8%), Cow milk (4–6%), Goat milk (3–4%) – easier to digest in adults.

Sucrose:

- Disaccharide, common "table sugar".
- Made of glucose and fructose.
- Too much can cause obesity, dental issues like cavities and plaque.



Proteins

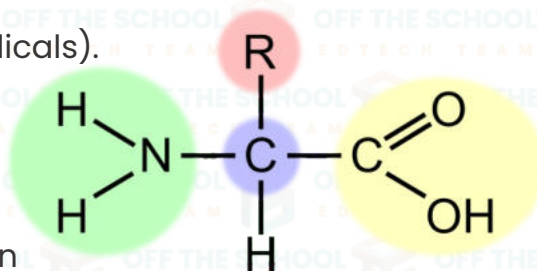
The term "protein" is derived from the Greek word "Proteios," meaning "first." Proteins are nitrogenous macromolecules that are present in all cells of living organisms. They hold a central role in the structure and functioning of living matter. Proteins consist mainly of carbon, hydrogen, nitrogen, and oxygen, with occasional presence of Sulphur and phosphorus. A protein is defined as a polymer, composed of simple units known as amino acids, linked together to form a larger molecule.

Amino acids as monomers or building blocks of protein:

Amino acids are building blocks of proteins. They are bi-functional compounds and contain basic amino ($-\text{NH}_2$) and acidic carboxyl group (COOH) groups. Up to twenty amino acids have been found in nature, ten are essential and remaining ten non-essential. Body can only synthesize non-essential ten amino acids. The general formula for amino acids is:

Where 'R' is the chain of carbon atoms (alkyl Radicals).

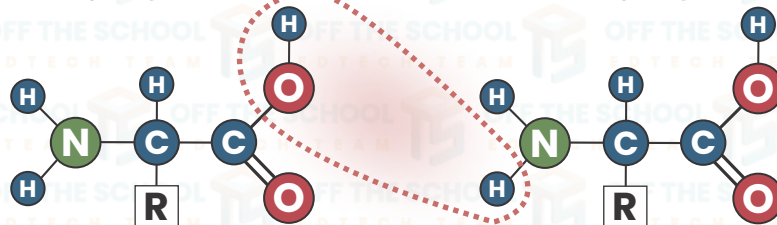
During the condensation of amino acids, $-\text{OH}$ (from carboxyl group) of one amino acid and H (from amino group) of another amino acid are combined and eliminated as water (H_2O) molecule. Thus, a new linkage is formed between two amino acid units known as **peptide linkage/ bond**. Due to this linkage protein is formed.



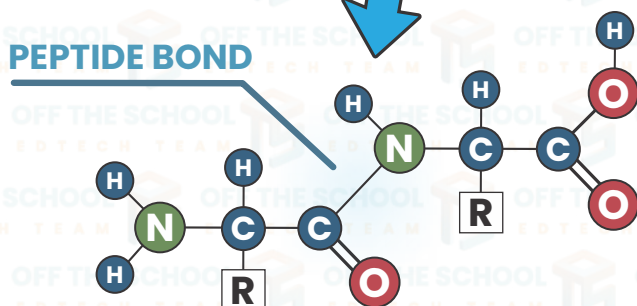
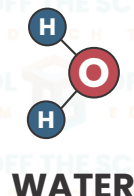
PEPTIDE BOND

AMINO ACID₁

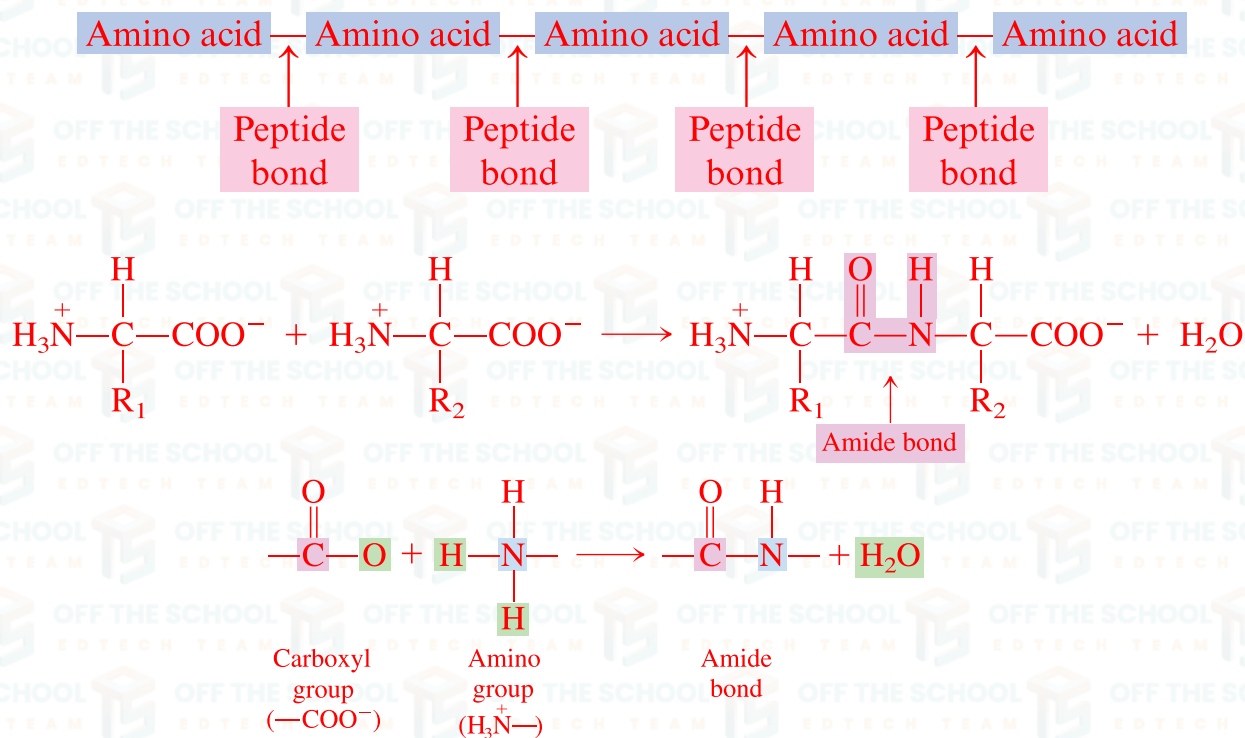
AMINO ACID₂



PEPTIDE BOND



DIPEPTIDE



A peptide bond is a covalent chemical bond that links together two amino acids within a protein or peptide molecule.

Classification of proteins:

Proteins can be classified into several groups on the bases of their structure and functions.

- 1. Catalytic proteins (Enzymes):** Serve as biological catalysts, increasing the rate of biochemical reactions. Examples include amylase, which breaks down starch into glucose, protease, which breaks down protein into amino acids, and lipase, which breaks down lipids into fatty acids.
- 2. Storage proteins:** Store nutrients or metal ions in plants or animals, such as Albumin, Globulin, and Casein.
- 3. Transport proteins:** Facilitate the movement of substances across cellular membranes and in the bloodstream, like hemoglobin.
- 4. Regulatory or hormonal proteins:** Play a crucial role in regulating bodily functions by transmitting signals between cells.

Classification of proteins based on their structure

classification of proteins based on their structure, broken down into four levels:

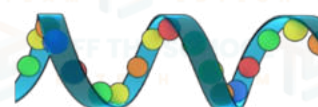
Primary Structure:

The linear sequence of amino acids in the protein chain. Determines the overall shape and function of the protein.



Secondary Structure:

Folding patterns in polypeptide chains due to amino acid interactions.



Common structures include alpha helix and beta sheets.

Stabilization occurs through hydrogen bonds between N-H and C=O groups of amino acids.

Tertiary Structure:

The three-dimensional arrangement of a protein molecule with a folded polypeptide chain.

Stability is due to interactions such as salt bridges, disulfide bridges, Van der Waals forces, and hydrogen bonds.

Example: Myoglobin exhibits a tertiary structure.



Quaternary Structure:

Complex proteins formed by multiple subunits interacting.

Example: Hemoglobin consists of four subunits and represents the quaternary structure.

**Sources and Uses of Proteins:**

The important sources of proteins are eggs, meat, pulses, beans, peas, and cheese etc.

Uses of Proteins:

1. Proteins are crucial for building and repairing muscles.
2. Proteins act as enzymes, facilitating chemical reactions in the body.
3. Some proteins serve as hormones, regulating various physiological processes.
4. Antibodies, a type of protein, help defend the body against infections and diseases.
5. Proteins aid in the transportation of molecules, such as oxygen in the blood, and the storage of important substances like iron and calcium.

Lipids:

Lipids are a heterogeneous group of organic compounds that are hydrophobic and soluble in organic solvents. They are essential components of cells, contributing to their structure and functioning.

Fatty Acids:

Fatty acids are a type of organic acid that serves as a building block for lipids. They are composed of a hydrocarbon chain with a carboxyl group ($-\text{COOH}$) at one end.

Fatty acids can be either saturated or unsaturated, depending on the presence or absence of double bonds in their hydrocarbon chain.

Example:

$\text{C}_{15}\text{H}_{31}\text{COOH}$ (Palmitic Acid).

$\text{C}_{17}\text{H}_{35}\text{COOH}$ (Stearic Acid).

Sources and Uses of Lipids:

1. **Animals:** Lipids are commonly found in animal-based foods such as meat, poultry, fish, eggs, and dairy products. Animal fats, like lard and butter, are rich sources of lipids.
2. **Plants:** Certain plant-based foods contain lipids, including oils derived from olives, sunflower seeds, soybeans, coconuts, avocados, and nuts such as almonds and walnuts.
3. **Foods:** Many processed foods, such as baked goods, snacks, and fried foods, contain added fats and oils that contribute to the lipid content.

Uses of Lipids:

1. Energy storage, providing a concentrated source of fuel for the body.
2. Structural component of cell membranes, maintaining cell integrity.
3. Insulation and protection of organs, regulating body temperature and providing cushioning.
4. Hormone production, serving as precursors for the synthesis of vital hormones.
5. Absorption of fat-soluble vitamins, facilitating their utilization in the body.



Classification of lipids:

Based on chemical composition, lipids are classified into simple lipids, compound lipids, and steroids.

1. Simple Lipids:

- Chemically esters, composed of fatty acids and alcohols, serving as energy sources.
- Subclassified into fats, oils, and waxes.
- Fats and oils are known as triglycerides or trimesters, formed by the condensation of three fatty acid molecules and one glycerol molecule.

2. Types of fatty acids:

- Saturated Fatty Acid:** Only single bonds between carbon atoms (e.g., Stearic Acid, Palmitic Acid).
- Unsaturated Fatty Acid:** At least one double bond between carbon atoms (e.g., Oleic Acid, Linoleic Acid).

3. Waxes:

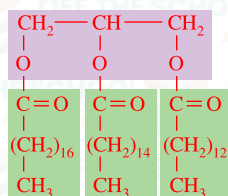
- Naturally occurring esters of long-chain fatty acids and alcohols.
- Water repellent, found on plant surfaces and used by bees to build honeycombs.

4. Compound Lipids:

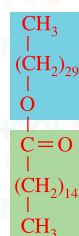
- Esters of glycerol with fatty acids and other compounds like carbohydrates, amino acids, and phosphoric acid.
- Classified into phospholipids and lipoproteins, with examples like LDL (low-density lipoprotein).

5. Steroids:

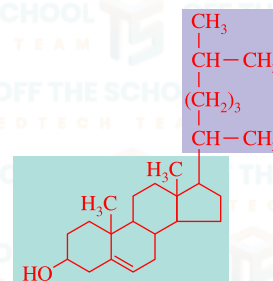
- Derived lipids with a structure of four interconnected carbon rings.
- Examples include cholesterol and cholic acid.



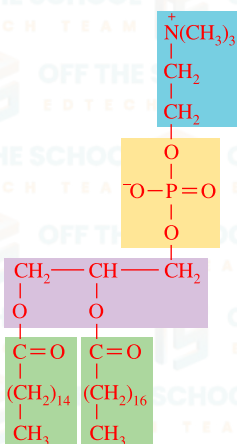
A fat



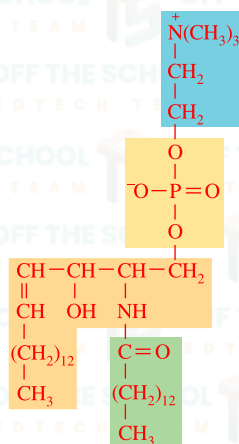
A biological wax



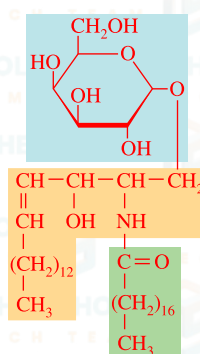
A steroid



A glycerophospholipid



A sphingophospholipid



A sphingoglycolipid

The structural formulas of these types of lipids illustrate the great structural diversity among lipids. The defining parameter for lipids is solubility rather than structure.

Physical Properties of lipids:

- Lipids have various physical states depending on their chemical nature and temperature (e.g., fats are solid, and waxes are semisolid at room temperature).
- They can be translucent or opaque and are insoluble in water but soluble in organic solvents like Bloor's reagent.
- Lipids have a low density, which allows them to float on water.
- The melting point of saturated lipids is higher than that of unsaturated lipids.
- As poor conductors of heat and electricity, lipids serve as insulators for animal bodies.

Chemical Properties of lipids:

- **Addition Reactions:** Oils can react with hydrogen in the presence of a nickel catalyst to form fats, converting unsaturated fatty acids into saturated ones (e.g., $\text{R-CH=CH-COOH} + \text{H}_2 \longrightarrow \text{R-CH}_2\text{-CH}_2\text{-COOH}$).
- **Saponification:** Fats and oils can be hydrolyzed with an alkali like sodium hydroxide to produce soap and glycerol; this process is called saponification.
- **Rancidity:** When animal fats are exposed to moist air, they undergo oxidation and hydrolysis reactions, leading to an unpleasant taste and odor, known as rancidity.

Minerals:

Minerals are inorganic substances vital for maintaining physical health and preventing diseases. Over 25 minerals are present in the body, with 16 recognized as essential. They are categorized as either macro minerals (needed in larger quantities) or micro minerals (needed in smaller quantities).

Sources of Important Minerals:

- **Calcium:** Found in milk, cheese, yogurt, and leafy greens.
- **Iron:** Present in red meat, poultry, seafood, beans, and lentils.
- **Zinc:** Available in beef, chicken, seafood, and beans.
- **Phosphorus:** Obtained from meat, fish, dairy products, nuts, seeds, and whole grains.

Biological Significance:

- **Iron:** Vital for oxygen transport, immune function, temperature regulation, and red cell production.
 - **Calcium:** Builds strong bones and teeth, helps muscle contraction and blood clotting, and supports the immune system.
 - **Phosphorus:** Crucial for healthy bones, teeth, DNA/RNA, protein synthesis, and maintaining heart rhythm.
 - **Zinc:** Important for growth, wound healing, nervous system function, immune system strength, insulin activity, and the senses of smell and taste.
- Deficiencies in these minerals can lead to serious health issues such as bone weakness, poor growth, and loss of appetite or taste.

